

Signal Processing Assignment 2

Gravitational Waves

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1 Analytical Problems

1.1 Problem 1

The following theorems describe how to obtain the initial and final steady state value of the input function $f[n]$ by analyzing its Z-transform $F(z)$.

problem 1

→ Initial value

$$F(z) = \sum_{n=-\infty}^{\infty} f[n] z^{-n}$$

$$\text{LTI syst: causality: } \sum_{n=0}^{\infty} f[n] z^{-n} = f[0] + f[1] z^{-1} + \dots + f[n] z^{-n}$$

$$\lim_{z \rightarrow \infty} F(z) = f[0]$$

→ Final value, steady state

$$z \{ f[n+1] - f[n] \} = \lim_{k \rightarrow \infty} \sum_{n=0}^k f[n+1] z^{-n} - \sum_{n=0}^k f[n] z^{-n}$$

$$= \lim_{k \rightarrow \infty} [-f[0] + f[1](1 - z^{-1}) + f[2](z^{-1} - z^{-2}) + \dots + f[k](z^{-k+1} - z^{-k}) + f[k+1] z^{-k}]$$

$$\lim_{k \rightarrow \infty} \{ z^{-k+1} - z^{-k} \} \rightarrow 0$$

$$\lim_{z \rightarrow 1} z \{ f[n+1] - f[n] \} = \lim_{z \rightarrow 1} \lim_{n \rightarrow \infty} [-f[0] + f[1](1 - z^{-1}) + f[2](z^{-1} - z^{-2}) + \dots + f[k](z^{-k+1} - z^{-k}) + f[k+1] z^{-k}]$$

$$\lim_{z \rightarrow 1} z(F(z) - f[0]) - F(z) = \lim_{n \rightarrow \infty} f[n+1] - f[0]$$

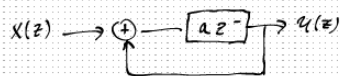
$$\lim_{z \rightarrow 1} (z-1)F(z) - z f[0] = \lim_{n \rightarrow \infty} f[n+1] - \lim_{n \rightarrow \infty} f[0]$$

$$\lim_{z \rightarrow 1} (z-1)F(z) - \cancel{f[0]} = \lim_{n \rightarrow \infty} f[n+1] - \cancel{f[0]}$$

$$\lim_{z \rightarrow 1} (z-1)F(z) = \lim_{n \rightarrow \infty} f[n+1] = f[\infty]$$

1.2 Problem 2

problem 2



$a \in \mathbb{R}$

(* in n space $\Rightarrow$ $\cdot$ in z space)

$$\begin{aligned} a) \quad u(z) &= a z^{-1} \{ x(z) + u(z) \} \\ u(z) &= a z^{-1} x(z) + a z^{-1} u(z) \quad \{ z^{-1} \} \\ y[n] &= a y[n-1] + a x[n-1] \end{aligned}$$

$$\begin{aligned} b) \quad (1 - a z^{-1}) u(z) &= a z^{-1} x(z) \\ u(z) &= \frac{a z^{-1}}{1 - a z^{-1}} x(z) \\ u(z) &= \frac{a}{z - a} x(z) \\ H(z) &= \frac{a}{z - a} \end{aligned}$$

denominator: $z - a$, $p = a$

$$\begin{aligned} y[n] &= z^{-1} \{ u(z) \} = y_p[n] + y_h[n] \\ \text{for bounded } x[n] \text{ is } y_p[n] \text{ bounded} \\ y_h[n] &= \sum_i k_i p_i^n = a^n \end{aligned}$$

bounded output when $|a| < 1$

$$\begin{aligned} c) \quad x[n] &= u[n] \\ x(z) &= z \{ u[n] \} = \sum_{n=0}^{\infty} 1 z^{-n} = \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1}, \quad |z| < 1 \Leftrightarrow |z| > 1 \\ u(z) &= H(z) \cdot x(z) = \frac{a}{z - a} \frac{z}{z - 1} = \frac{a z}{(z - 1)(z - a)} \quad |z| > 1 \end{aligned}$$

d) Init and Final value theorem for $H(z)$

$$h[0] = \lim_{z \rightarrow \infty} H(z) = \lim_{z \rightarrow \infty} \frac{a}{z - a} = 0 \quad \text{indep of } a$$

$$h[\infty] = \lim_{z \rightarrow 1} (z - 1) H(z) = \lim_{z \rightarrow 1} \frac{(z - 1) a}{(z - 1)(z - a)} = 0$$

$$\text{if } a \neq 1: h[\infty] = \frac{(z - 1) 1}{(z - 1)} = 1$$

2 Gravitational Waves

The following section describes the procedure and results for obtaining and processing the Gravitational Wave signal caused by two merging black holes. We use the recorded raw data GW150914 from LIGO. Signal processing will be done in Python using the extended libraries Numpy, Scipy (scipy.signal) and Matplotlib.

2.1 Raw Signal

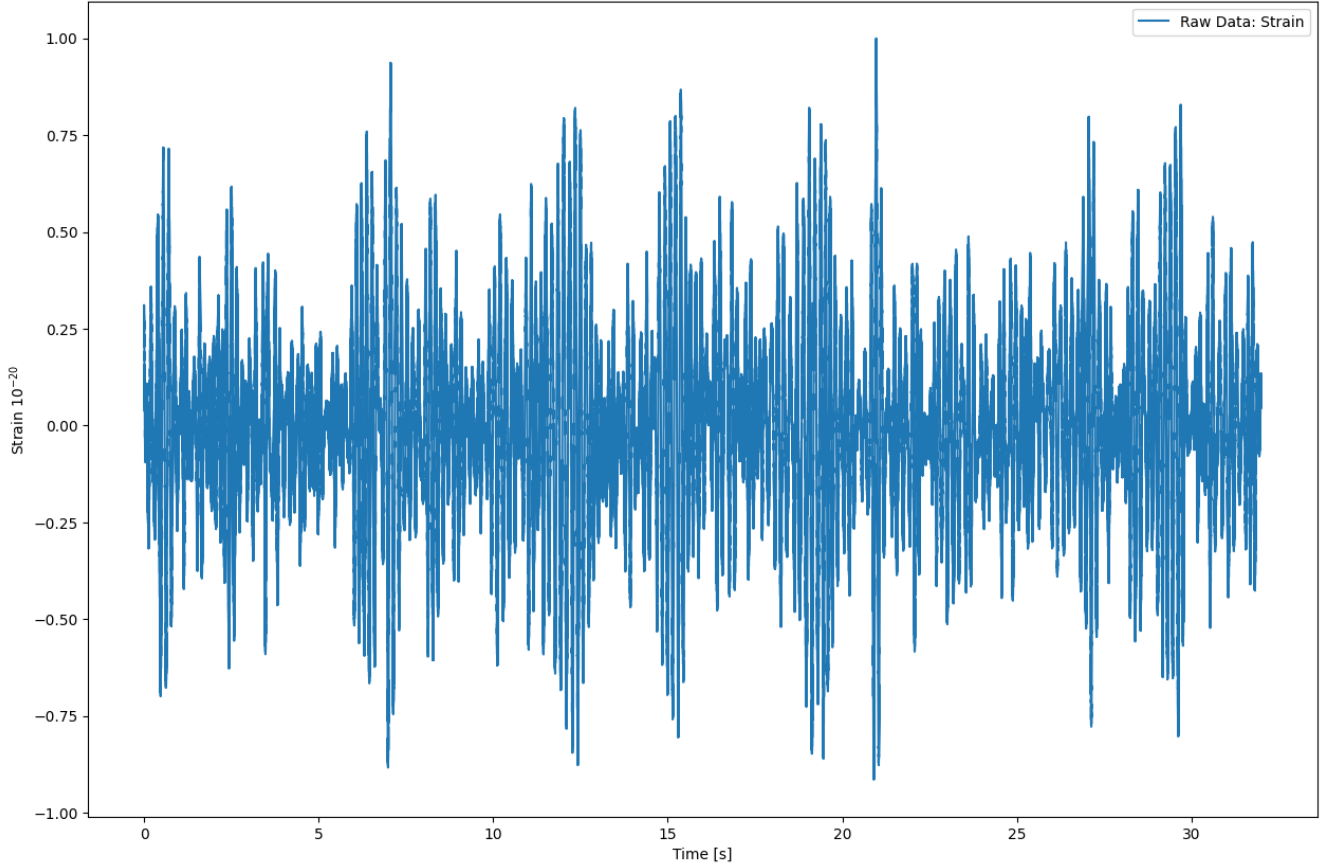


Figure 1: Raw strain data retrieved from GW150914, LIGO. The strain is recorder in magnitude of roughly -20 . To avoid numerical errors we must normalize the data set by dividing with the smallest recorded value.

2.2 Fourier Transform of the Raw Signal

Before we start filtering, we need to estimate the sampling frequency F_s using the formula:

$$F_s = \frac{N}{t_{\text{end}} - t_{\text{start}}}$$

where N is the number of data points. Utilizing **Numpy.fft.fftfreq** for the Fourier transform, we obtain the frequencies present in the unfiltered signal.

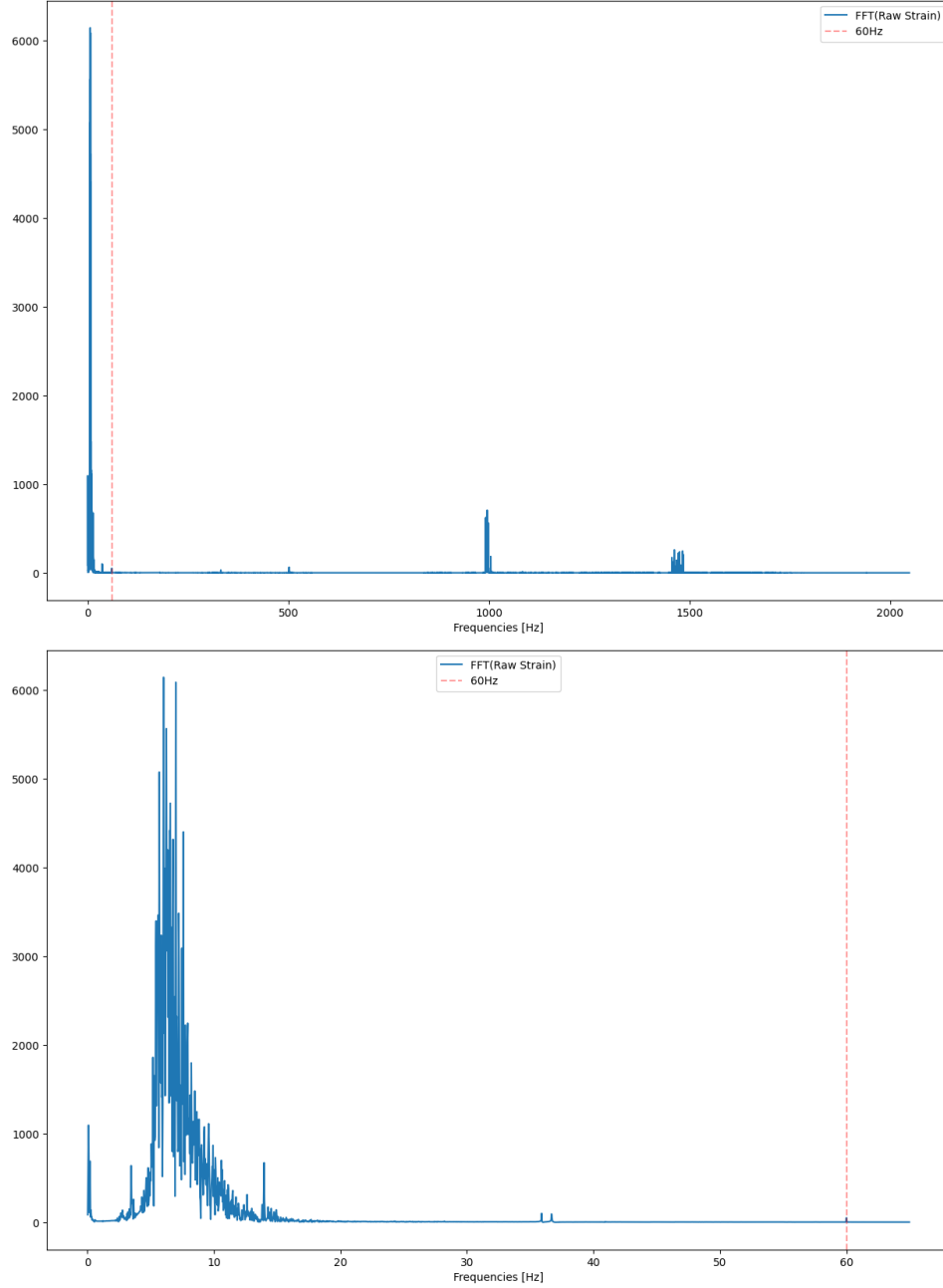


Figure 2: Fourier transform of the raw signal. Some high-frequency noise is present. The main focus will be frequencies between 50Hz and 250Hz. We highlight the 60Hz spike that attributes to one of the AC current modes.

2.3 Notch Filtering the AC current

We implement a notch filter to remove the noise from the AC current located at $60n, n \in [1, 2, 3]$. We implement `iirnotch` from `scipy.signal` and specify the previously mentioned sampling frequency f_s and the quality factor $Q = \frac{f_n}{BW} = 15$. Here BW is the bandwidth and f_n is the notch frequency.

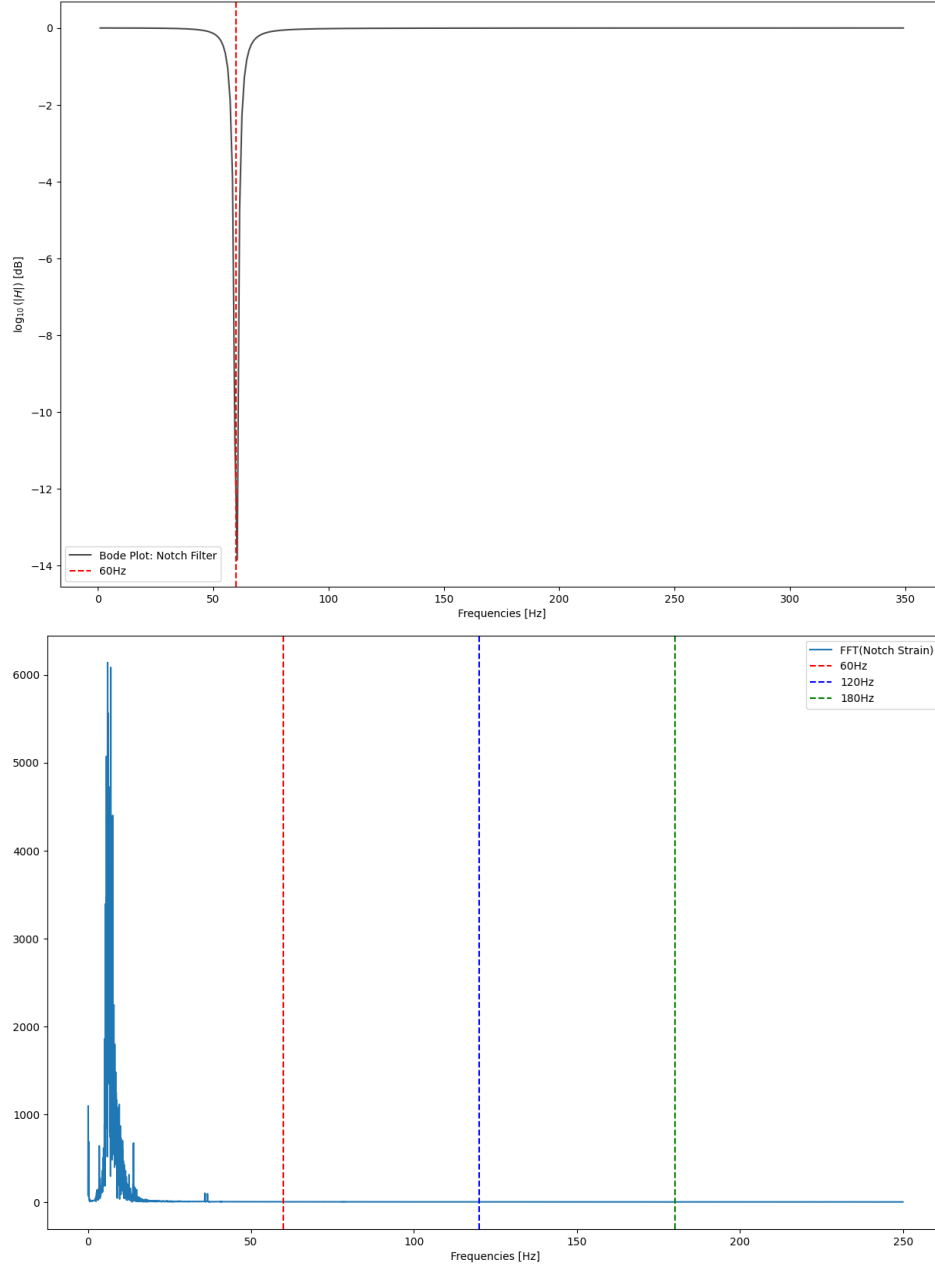


Figure 3: Top graph shows the bode plot of a notch filter with notch frequency 60Hz and quality factor 15 (high value). Bottom graph shows the Fourier transform of the filtered signal, with all AC frequencies removed. Note how the small 60Hz spike is now gone.

2.4 Band Pass Filter

Now we filter away all the frequencies outside 50Hz and 250Hz interval. In the program we use the Butterworth filter of 4th order. We sacrifice faster fall rate for uniform pass band, which for such a sensitive experiment is much more valuable.

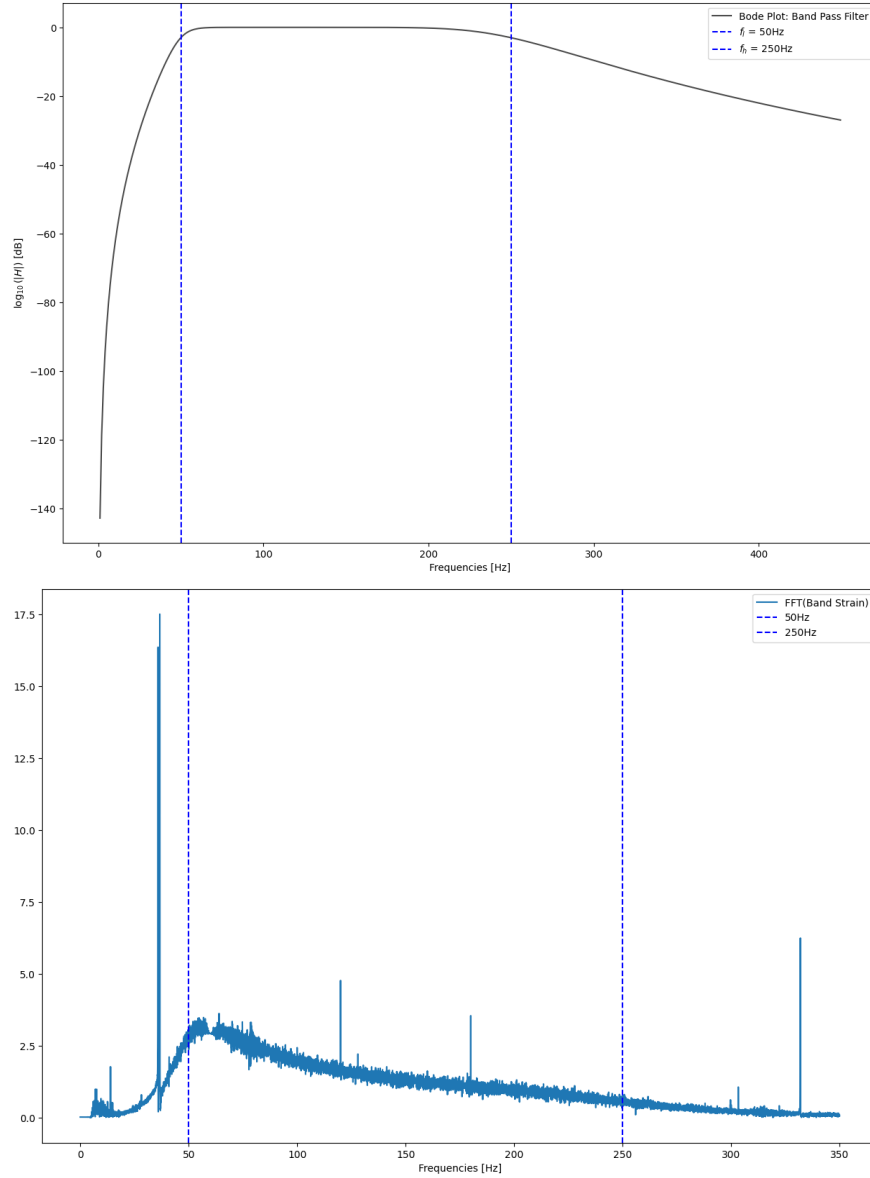


Figure 4: Top graph shows the bode plot of a 4th order Butterworth band pass filter with cutoff frequencies 50Hz and 250Hz. The weak fall rate after 250Hz might be the **signal.freqz** function having representation issues. Bottom graph shows the Fourier transform of the filtered signal. Notice the scaling on the y axis compared to the previous plots. The signal to noise ratio for the raw data was clearly large. The weak gravitational signal gets affected by a number of background noise sources.

2.5 Filtered Signal

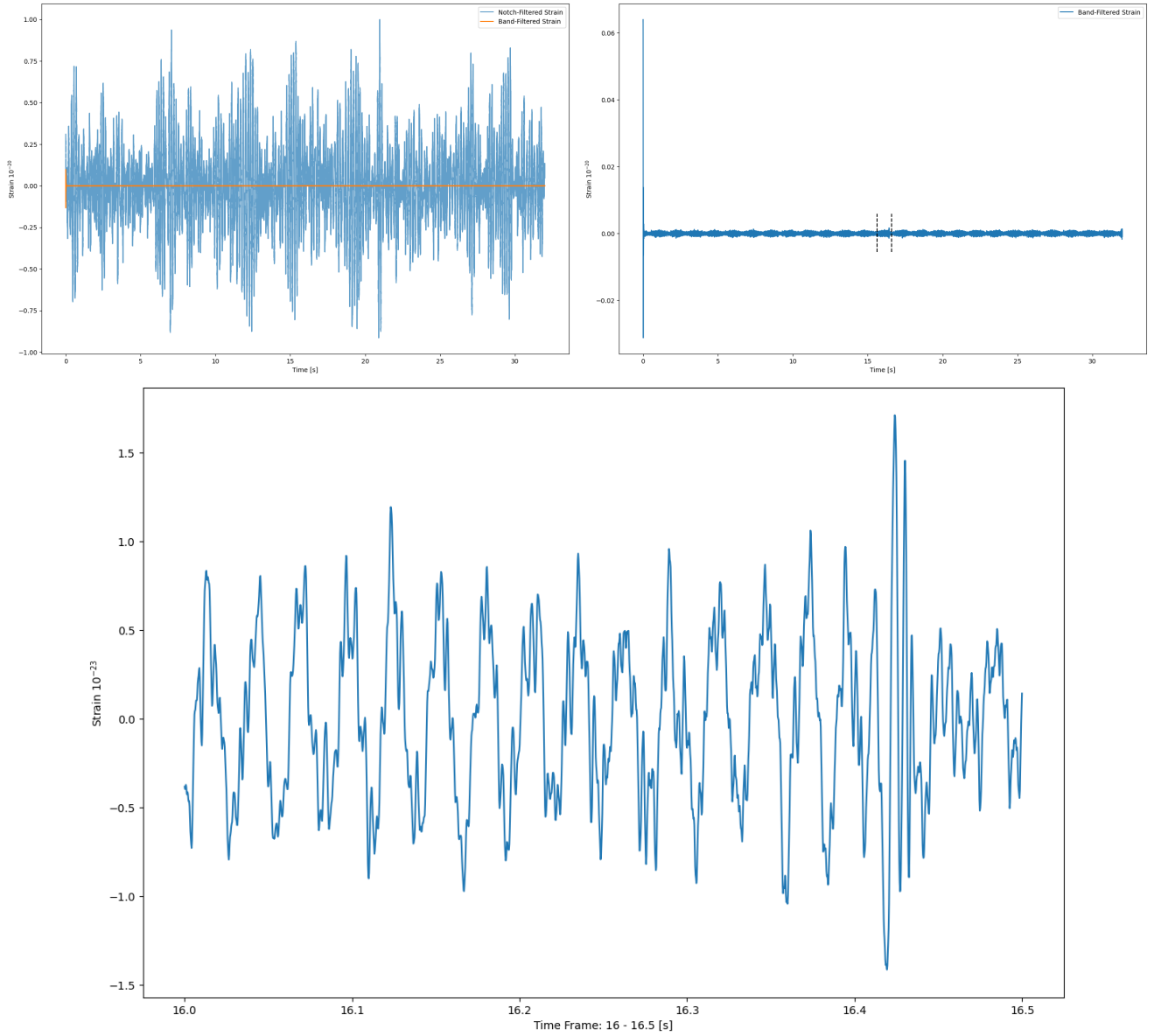


Figure 5: Top graph show the result of filtering with the notch and band pass filters. Again notice the influence of noise outside the 50Hz to 250Hz interval.

In the middle, the filtered signal is plotted alone. The black dotted lines highlight the region between 16 and 16.5 seconds where there is unprecedented fluctuation (this is the black hole merging).

Bottom graph zooms into this region revealing the merge. This signal has now significantly fewer disturbances and is comparable to the signal published by LIGO.